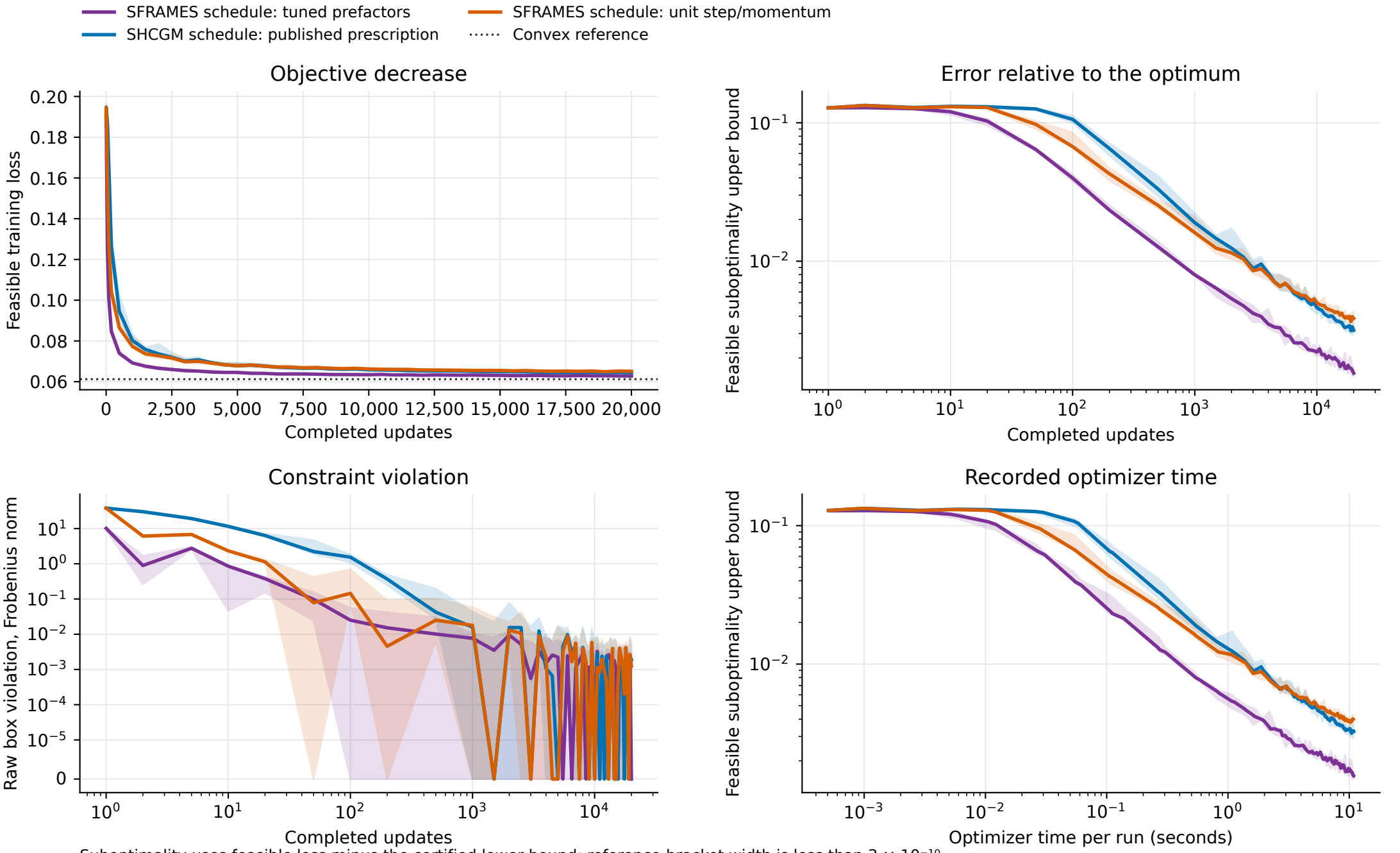
Matrix completion: schedule comparison

50 × 50 · FP64 · Batch 64 · Median and range of five fresh seeds · Current feasible outputs

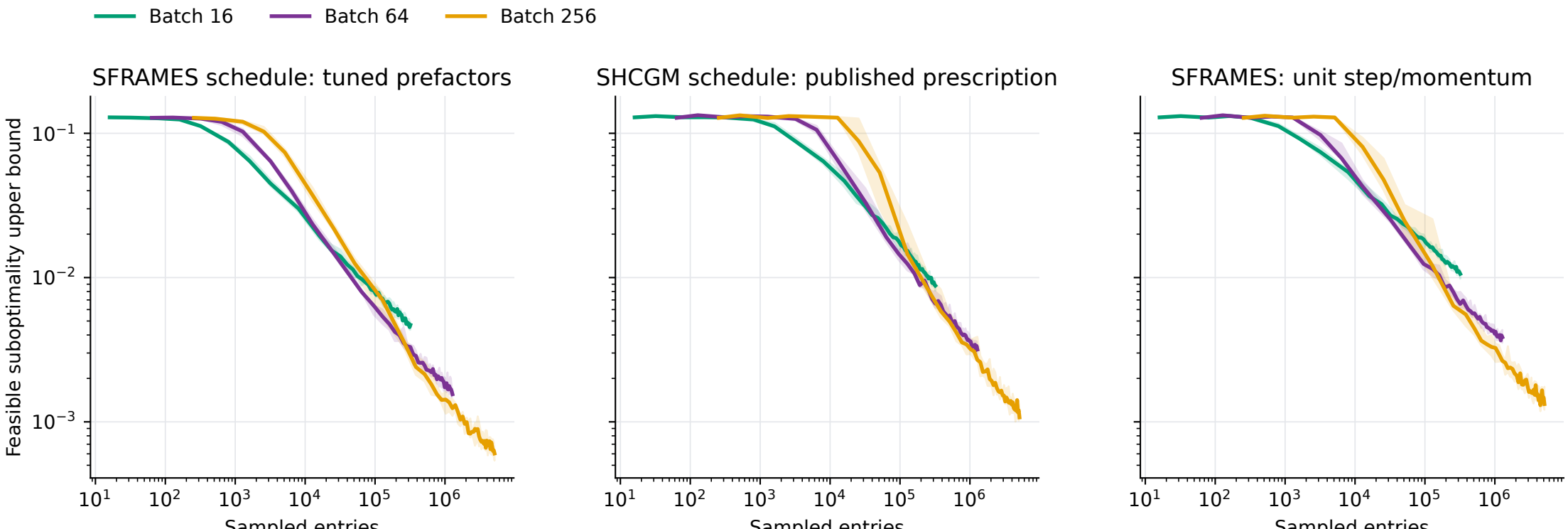


Suboptimality uses feasible loss minus the certified lower bound; reference-bracket width is less than 3×10^{-10}.

Timing excludes diagnostics and I/O. Time bands interpolate only within shared observed support; four pinned CPU workers ran concurrently.

Matrix completion: batch size and sampled observations

Each point counts actual sampled training entries, including repeats · Five fresh seeds per batch



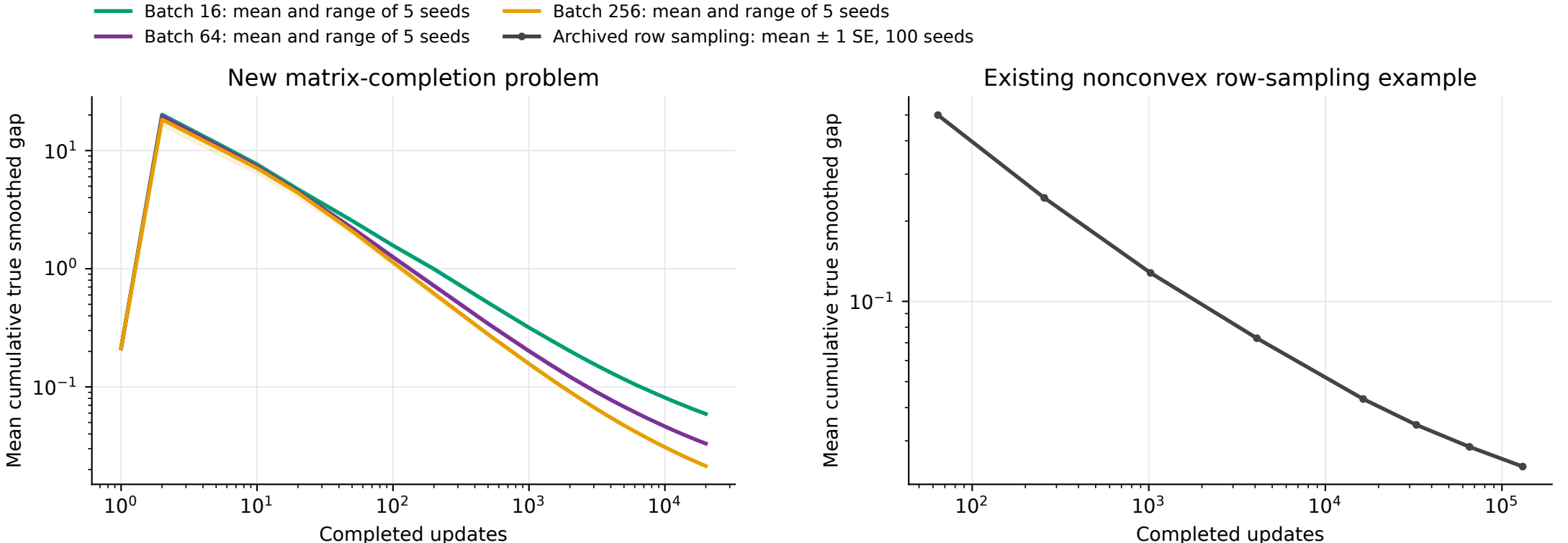
All runs end at 20,000 updates: batch 16 = 320,000 draws; batch 64 = 1,280,000; batch 256 = 5,120,000.

Prefactors were selected at batch 64 and reused unchanged at 16 and 256. Curves stop at their actual observation budgets.

Shaded regions are observed seed ranges, not confidence intervals. Loss is evaluated after radial feasibility repair, used only for diagnostics.

Theorem statistic: true averaged smoothed FW gap

Every pre-update iterate contributes its own exact gap. Expectations estimated across independent seeds



Left: SFRAMES with $\gamma_0=\rho_0=1$ and selected β_0; mean of per-run arithmetic averages over all $k=0,\ldots,K-1$.

Right: archived 1,024-variable construction, batch1. Its global optimum and full gradient are available analytically.

These are different problems and are not a solver ranking. Declining empirical gaps illustrate the theorem; they do not prove its asymptotic rate.